Agustín Nolasco

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Education

2024–Present **PhD. in Computer Science** in National University of Córdoba, FAMAF. Córdoba. Ongoing.

2019–2024 **Licentiate in Computer Science** in National University of Río Cuarto. Córdoba.
Final Average Score: 9.88 out of 10.
Thesis title: *Inference of Equivalent Method Sequences for Metamorphic Testing*.

2019–2021 **Computing Analyst** in National University of Río Cuarto. Córdoba.
Final Average Score: 9.81 out of 10.

Teaching Experience

2024 – Present **Teaching Assistant,** National University of Río Cuarto, Argentina
Courses: Logics and Problem Solving – Compilers – Design of Algorithms – Algorithms and Data Structures

2023 – Present **Grader,** Guangdong Technion – Israel Institute of Technology, China
Courses: Seminars in Computer Science – Introduction to Computer Science – Introduction to System Programming

2022 – 2024 **Student Teaching Assistant,** National University of Río Cuarto, Argentina
Courses: Advanced Programming – Design of Algorithms

Industrial Experience

Apr. 2024 – Aug. 2024 **QA Tester,** Oliver Informatics, Río Cuarto, Córdoba, Argentina

Research Publications

Journal Articles

1. A. Nolasco, F. Molina, R. Degiovanni, *et al.*, "Abstraction-aware inference of metamorphic relations," *Proc. ACM Softw. Eng.*, vol. 1, no. FSE, Jul. 2024. DOI: 10.1145/3643747.

Conference Proceedings

1. A. Balestra, A. Nolasco, F. Molina, D. Garbervetsky, R. Degiovanni, and N. Aguirre, "Improving dynamic specification inference with llm-generated counterexamples," in *Proceedings of the 19th IEEE International Conference on Software Testing, Verification and Validation (ICST)*, To appear in ICST 2026, 2026.

Preprints

1. R. Degiovanni, F. Molina, A. Nolasco, N. Aguirre, and M. Papadakis, *Specification inference for evolving systems*, Pre-print, 2023. arXiv: 2301.12403 [cs.SE].

Courses

Apr. 2025 **Parallel Computing**
dictated by Dr. Nicolás Wolovick, FAMAF, National University of Córdoba
Ongoing.

Aug. 2024 **Text Mining**
dictated by Dra. Laura Alonso Alemany, FAMAF, National University of Córdoba
Ongoing.

Neural Networks
dictated by Dr. Francisco Tamarit, FAMAF, National University of Córdoba
Final Score: 10 out of 10

May. 2025 **Introduction to Software Testing**
dictated by Dr. Andrea Bombarda
Ongoing.

Mar. 2025 **Advances in Automatic Software Testing**
dictated by Dr. Mike Papadakis, Río 2025 Summer School
Final Score: 9 out of 10

Introduction to Input Grammar Fuzzing
dictated by Dr. Andreas Zeller, Río 2025 Summer School
Final Score: 10 out of 10

Analysis of mobile apps for quality and trustworthiness
dictated by Dr. Alessandra Gorla, Río 2025 Summer School
Final Score: 10 out of 10

Feb. 2024 **Dynamic Modal Logics**
dictated by Dr. Raul Fervari and Dr. Carlos Areces, Río 2024 Summer School
Final Score: 6 out of 10

Verification of Fault Trees
dictated by Dr. Joost-Pieter Katoen, Río 2024 Summer School

Aug. 2023 **Advanced Data Structures**
dictated by Dr. Conrado Martínez, ECI 2023
Final Score: 10 out of 10

Fuzzing for Compiler Testing
dictated by Dr. Guillermo Polito, ECI 2023
Final Score: 9 out of 10

Feb. 2023 **Modeling and Analyzing Concurrent Systems**
dictated by Dr. Thomas Noll, Río 2023 Summer School
Final Score: 8 out of 10

Formal Approaches to Decision-Making under Uncertainty
dictated by Dr. Arnd Hartmanns, Río 2023 Summer School

Aug. 2022 **Introduction to APIs**
dictated by Dr. Guido De Caso, ECI 2022
Final Score: 8 out of 10

Satellite Images Processing
dictated by Dr. Rafael Grimson, ECI 2022
Final Score: 7 out of 10

Oct. 2019 **Software Testing**
dictated by Dr. Nazareno Aguirre, CACIC 2019
Final Score: 10 out of 10

Courses (continued)

2011–2016 **English Academy**
Superior Institute of English Language, Villa Mercedes, San Luis

Research Projects

Dec. 2021 – Present **Optimizations to Symbolic Execution and Applications**
PICT 2019
Director: Dr. Nazareno Aguirre
Position: Collaborator
Funded by: National Agency for the Promotion of Research, Technological Development, and Innovation, through FONCyT.

Jan. 2020 – Present **Application of Constraint Solving Techniques to Software Validation and Verification**
PPI 2020-2022
Director: Dr. Nazareno Aguirre
Position: Collaborator
Funded by: Secretariat of Science and Technology, National University of Río Cuarto.

Research Grants and Scholarships

Ago. 2024 – Present **PhD. Scholarship**. National Scientific and Technical Research Council (CONICET).

Sept. 2022 – Aug. 2023 **EVC-CIN Scholarship 2021**. Stimulation Scholarship for Scientific Vocations.

Honors

2023 – 2024 **First Major National Flag Escort**, National University of Río Cuarto.

Talks and Presentations

2026 **Improving Dynamic Specification Inference with LLM-Generated Counterexamples**
Invited talk, Argentine Workshop on Fundamentals for the Automatic Analysis and Construction of Software (FACAS) 2026, La Falda, Argentina.

2022 **Abstraction-aware Inference of Metamorphic Relations**
Invited talk, Argentine Workshop on Fundamentals for the Automatic Analysis and Construction of Software (FACAS) 2022, La Falda, Argentina.

Supervision

2025 – Present **Co-advisor**, Undergraduate thesis,
LLM-based techniques for improving dynamic specification inference.
By Agustín Balestra.

2024 – 2025 **Co-advisor**, Undergraduate thesis,
Construction and Integration of Automatic Generators of Complex Inputs in Libraries Supporting Property-Based Testing.
By Augusto Fungo.